



TEST

1

Which of the following is an invalid class declaration in C++:

- a) class A { int x; };
- b) class B { };
- c) public class A { };
- d) static class B { };

2

What will be the output for the following C++ source code:

```
#include <iostream>
#include <stdio.h>
using namespace std;
class Utils {
public:
    static short method(char* array, short length) {
        short c = 0;
        for (short i = 0; i < length; i++) {
            if (array[i] != 0) {
                break;
            }
            else {
                c++;
            }
        }
        return c;
    }
};
int main(int argc, char** argv) {
    char v1[8] = { 0, 0, 1, 3, 5, 0, 0, 0 };
    short c = Utils::method(v1, 8);
    for (short i = c; i < sizeof(v1)/sizeof(v1[0]); i++) {
        printf(" %02X,", v1[i]);
    }
    return 0;
}
```

- a) There is no output because the source code has the compilation errors.
- b) 01, 03, 05
- c) 00, 00, 01, 03, 05, 00, 00, 00,
- d) 01, 03, 05, 00, 00, 00,

3

Which operator should be overloaded in the following C++ code to make the program error free?



```
#include <iostream>
#include <string>
using namespace std;
class Box{
    int capacity;
public:
    Box(){}
    Box(double capacity){
        this->capacity = capacity;
    }
};
int main(int argc, char const *argv[])
{
    Box b1(10);
    Box b2 = Box(14);
    if(b1 == b2){
        cout<<"Equal";
    }
    else{
        cout<<"Not Equal";
    }
    return 0;
}
```

- a) +
- b) =
- c) ==
- d) ()

4

What is the output of the following C program?

```
#include <stdio.h>
void xyz(int p1, int *p2) {
    ++p1;
    ++*p2;
    printf("%d%d",p1,*p2);
}
void main() {
    int a=10;
    xyz(a++,&a);
    xyz(a++,&a);
    printf("%d",a);
}
```

- a) 1011121313
- b) 1112131314
- c) 1011121213
- d) 1112131414

5

What is the output of following C program?



```
#include <stdio.h>
void main(){
    int i;
    for(i=1;i++<=1;i++)
        for(i++;i++<=6;i++)
            i++;
    printf("%d",i);
}
```

- a) 12
- b) 11
- c) 13
- d) none of these

6

What will be the output of the following C code:

```
int v[] = {1,2,3,4,5}, n = 5;
for (int i = 0; i < n / 2; i++) {
    v[i] ^= v[n - i - 1];
    v[n - i - 1] ^= v[i];
    v[i] ^= v[n - i - 1];
}
for (int i = 0; i < n; i++) {
    printf("%d ", v[i]);
}
```

- a) 5 4 3 2 1
- b) 1 2 3 4 5
- c) 0 0 0 0 0
- d) None of the mentioned

7

What is the output of following C program?

```
#include <stdio.h>
void main(){
    int a=1;
    while(a++<=1)
        while(a++<=2);
    printf("%d",a);
}
```

- a) 5
- b) 4
- c) 1
- d) 6

8

What will be the output for the following Java source code:

```
package eu.ase;
import java.io.BufferedReader;
import java.io.BufferedOutputStream;
```



```
import java.io.DataInputStream;
import java.io.DataOutputStream;
import java.io.EOFException;
import java.io.FileInputStream;
import java.io.FileOutputStream;
import java.io.IOException;
public class Test {
    public static void main(String[] args) {
        double[] frequency = {12, 3, 21, 24};
        int[] values = {7, 9, 10, 11};
        try {
            DataOutputStream out = new DataOutputStream(new
BufferedOutputStream(new FileOutputStream("test.txt")));
            for (int i = 0; i < frequency.length; i++) {
                out.writeDouble(frequency[i]);
                out.writeInt(values[i]);
            }
            out.flush();
            out.close();
            DataInputStream in = new DataInputStream(new BufferedInputStream(new
FileInputStream("test.txt")));
            try {
                while (true) {
                    double freqv = in.readDouble();
                    int val = in.readInt();
                    System.out.format("Value %d - %.2f frequency ", val, freqv);
                }
            } catch (EOFException e) {in.close();}
        } catch (IOException ioe) {ioe.printStackTrace();}
    }
}
```

- a) Value 7 - 12.00 frequency Value 9 - 3.00 frequency Value 10 - 21.00 frequency Value 11 - 24.00 frequency
- b) The example generates compiler errors
- c) The example generates runtime exceptions
- d) Value 7 - 24.00 frequency Value 9 - 21.00 frequency Value 10 - 3.00 frequency Value 11 - 12.00 frequency

9

Which of the below is not an implementation of List interface?

- a) ArrayList
- b) Stack
- c) LinkedList
- d) SessionList

10

In the following C++ code how many times the string "A's constructor called" will be printed?

```
#include <iostream>
```



```
#include <string>
using namespace std;
class A {
    int a;
public:
    A() {
        cout<<"A's constructor called";
    }
};
class B {
    static A a;
public:
    B() {
        cout<<"B's constructor called";
    }
    static A get() {
        return a;
    }
};
A B::a;
int main(int argc, char const *argv[])
{
    B b;
    A a1 = b.get();
    A a2 = b.get();
    A a3 = b.get();
}
```

- a) 1
- b) 4
- c) 2
- d) 3

11

What will be the result of executing the following C++ code:

```
#include <iostream>
using namespace std;
class rect
{
public:
    int area(int x=4,int y=3)
    {
        return (x * y);
    }
};
int area(int x=4,int y=3) { return 2*x*y; }
int main()
{
    rect rect;
    cout << area();
}
```



```
return 0;
```

```
}
```

- a) Shows 12
- b) Shows 16
- c) None of the mentioned
- d) Shows 24

12

What is the output of the following C++ code?

```
#include <iostream>
#include <string>
using namespace std;
class Box{
    int capacity;
public:
    Box(){}
    Box(double capacity){
        this->capacity = capacity;
    }
    bool operator<(Box b){
        return b.capacity < this->capacity? true : false;
    }
};

int main(int argc, char const *argv[])
{
    Box b1(10);
    Box b2 = Box(14);
    if(b1 < b2){
        cout<<"B1's capacity is smaller";
    }
    else{
        cout<<"B2's capacity is smaller";
    }
    return 0;
}
```

- a) B1's capacity is smaller
- b) B2's capacity is smaller
- c) Error
- d) Segmentation fault

13

What is the output of this C++ program?

```
#include <iostream>
using namespace std;
class MyInterface
{
public:
```



```

virtual void Display() = 0;
};
class Class1 : public MyInterface
{
    public:
    void Display()
    {
        int a = 5;
        cout << a;
    }
};
class Class2 : public MyInterface
{
    public:
    void Display()
    {
        cout << " 5" << endl;
    }
};
int main()
{
    Class1 obj1;
    obj1.Display();
    Class2 obj2;
    obj2.Display();
    return 0;
}

```

- a) 5
- b) 10
- c) None of the mentioned
- d) 5 5

14

What is the output of the following Java program:

```

class B {
    void print() {
        System.out.println("Class B");
    }
}
class A extends B {
    @Override
    public void print() {
        System.out.println("Class A");
    }
}
public class S4405 {
    public static void main(String[] args) {
        B b = new A();
        b.print();
    }
}

```



}

- a) Class B
- b) Class A
- c) Compilation error message
- d) Runtime error message

15

What will happen when you attempt to compile and run the following Java code?

```
public class JavaS1 {  
    static {  
        int a = 10;  
    }  
    static int a, b;  
    public static void main(String args[]) {  
        a--;  
        myMethod();  
        System.out.println(a + b + a++);  
    }  
    public static void myMethod() {  
        b = a++ + ++a;  
    }  
}
```

- a) Compile-time error
- b) prints : 3
- c) prints : 2
- d) prints : 43

16

What will be the output for the following Java source code:

```
package eu.ase;  
class Container {  
    private String name;  
    private static Container instance = null;  
    private Container() {  
        this.name = "NA";  
    }  
    public static Container getInstance() {  
        if (instance == null) {  
            instance = new Container();  
        }  
        return instance;  
    }  
    public void setName(String x) {  
        this.name = x;  
    }  
    public String getName() {  
        return this.name;  
    }  
}
```




```
public class Test {  
    public static void main(String[] args) {  
        Container ob1 = Container.getInstance();  
        Container ob2 = Container.getInstance();  
        ob1.setName("Container 1");  
        ob2.setName("Container 2");  
        System.out.println("s1=" + ob1.getName() + ", s2=" + ob2.getName());  
    }  
}
```

- a) s1=Container 1, s2=Container 2
- b) The example generates compiler errors
- c) s1=Container 2, s2=Container 2
- d) s1=Container 1, s2=Container 1

17

How many times the below loop will get executed?

```
#include <stdio.h>  
void main() {  
    int i,j;  
    i = 10;  
    for (j=i==10 ; j<=10 ; j++) {  
        printf("\n%d", j);  
    }  
}
```

- a) 1
- b) Compilation Error
- c) 11
- d) 10

18

The following C++ program:

```
#include<iostream>  
using namespace std;  
class Person { public: void Show() { cout << "a person"; } };  
class Student : public Person { public: void Show() { cout << "a student"; } };  
void main()  
{  
    Person* p = new Student();  
    p->Show();  
    delete p;  
}
```

- a) does not compile because it contains syntax errors
- b) outputs "a student"
- c) outputs "a person"
- d) does not compile because we can't assign a Student to a Person

19

What is the output of the following C++ program?



```
#include <iostream>
using namespace std;
void swap(int &a, int &b);
int main()
{
    int a = 5, b = 10;
    swap(a, b);
    cout << " In main " << a << b;
    return 0;
}
void swap(int &a, int &b)
{
    int temp;
    temp = a;
    a = b;
    b = temp;
    cout << "In swap " << a << b;
}
```

- a) In swap 105 In main 105
- b) In swap 105 In main 510
- c) In swap 510 In main 105
- d) None of the mentioned

20

What will be the output for the following C++ source code:

```
#include <iostream>
#include <stdio.h>
using namespace std;
class Utils {
public:
    static void method(const char* array, short length, short* startOff, short* stopOff,
short* newLength) {
        int i = 0;
        *startOff = 0; *stopOff = length; *newLength = length;
        while (array[i] == 0x00)
            i++;
        *startOff = i;
        while (array[length - 1] == 0x00)
            length--;
        if (length > i)
            * newLength = (length - i);
        else
            *newLength = 0;
        *stopOff = length;
    }
};
int main(int argc, char** argv) {
    short startOff = 0; short stopOff = 0; short newLength = 0;
    char v1[8] = { 0, 0, 1, 2, 0, 4, 0, 0 };
    Utils::method(v1, 8, &startOff, &stopOff, &newLength);
```



```
printf("\n startOff = %d, stopOff = %d, newLength = %d ", startOff, stopOff,
newLength);
for (short i = startOff; i < stopOff; i++) {
    printf(" %02X", v1[i]);
}
for (short i = 0; i < newLength; i++) {
    printf(" %02X", (&v1[startOff])[i]);
}
return 0;
}
```

- a) There is no output because the source code has the compilation errors.
b) startOff = 2, stopOff = 6, newLength = 4 00, 00, 01, 02, 00, 04, 01, 02, 00, 04,
c) startOff = 2, stopOff = 6, newLength = 4 01, 02, 00, 04, 01, 02, 00, 04,
d) startOff = 2, stopOff = 6, newLength = 4 01, 02, 00, 04, 00, 00, 01, 02, 00, 04,

21

Which statement is not true in Java language?

- a) A public member of a class can be accessed in all the packages
b) A protected member of a class can be accessed from its derived class
c) A private member of a class cannot be accessed from its derived class
d) A private member of a class cannot be accessed by the methods of the same class

22

What is the output of the following C++ program?

```
#include<iostream>
using namespace std;
class X {
    int m;
public:
    X() : m(10) {}
    X(int mm): m(mm) {}
    int getm() { return m; }
};
class Y : public X
{
    int n;
public:
    Y(int nn) : n(nn) {}
    int getn() { return n; }
};
int main()
{
    Y yobj(100);
    cout << yobj.getm() << " " << yobj.getn() << endl;
}
```



- a) 100 10
- b) 10 100
- c) 10 10
- d) 100 100

23

What will be the result of executing the following C code:

```
#include <stdio.h>
void main() {
    int i;
    for(i=20, i=10; i<=20; i++) {
        printf("\n %d", i);
    }
}
```

- a) Shows 11
- b) Runtime Error
- c) Shows 1
- d) Compilation Error

24

What will be displayed by the following C++ program?

```
#include <iostream>
using namespace std;
class A
{
public:
    int &g()
    {
        static int i=10;
        return i;
    }
};
class B
{
public:
    A &f()
    {
        static A o;
        return o;
    }
};
void main()
{
    B b;
    cout<<b.f().g();
}
```

- a) it displays 0
- b) it incorrectly addresses a class member



- c) it incorrectly uses the object reference
- d) it displays 10

25

Which of the below is not an implementation of Java Set interface?

- a) SortedSet
- b) TreeSet
- c) HashSet
- d) LinkedHashSet

26

What is the output of following program?

```
#include <stdio.h>
```

```
void abc(int a) {  
    printf("%d ", ++a);  
}
```

```
void main(){  
    int a=10;  
    abc(++a);  
    abc(a++);  
    printf("%d", a);  
}
```

- a) 12 13 14
- b) 13 14 15
- c) 14 14 14
- d) 12 12 12

27

In C++, the constructor is executed when:

- a) a class is declared
- b) an object is used
- c) an object is created
- d) an object goes out of scope

28

The following program:

```
#include<iostream>  
using namespace std;  
class Person  
{  
public:  
    Person() { id = counter++; }  
    void Show() { cout << "Person #" << id << endl; }  
private:  
    static int counter;  
    int id;  
};  
int Person::counter = 1;
```



```
void main()
```

```
{  
    Person p1, p2, p3;  
    p2.Show();  
}
```

- a) outputs "Person #0"
- b) outputs "Person #2"
- c) outputs "Person #1"
- d) does not compile because it uses the variable counter before initialization

29

Which one from the next Java statements is correct:

- a) abstract classes can contain non-abstract methods and instance variables
- b) interfaces can contain instance variables
- c) abstract classes can be instantiated in objects
- d) none of these answers

30

Regarding virtual methods which statement is correct in Java:

- a) a method is virtual if it has the signature of the method within an interface
- b) a method is virtual if the 'virtual' keyword is used
- c) none of these answers
- d) all non-static methods are virtual by default and design

Notice:

Each right answer has three points. Maximum number of points is 90.

Seria CSIE 4 Data 26.07.2019

Nr.

a

b

C

d

3	punkte				
3	2	1	○	○	●
3	2	2	○	○	○
3	2	3	○	○	●
3	2	4	○	○	○
3	1	5	●	○	○
3	1	6	●	○	○
3	5	7	●	○	○
3	2	8	●	○	○
3	2	9	○	○	○
8	2	10	●	○	○
3	2	11	○	○	○
3	5	12	○	●	○
3	5	13	○	○	○
3	2	14	○	●	○
3	2	15	○	○	●
3	5	16	○	○	●
3	2	17	○	○	○
3	2	18	○	○	●
3	2	19	●	○	○
3	5	20	○	○	●
3	2	21	○	○	○
3	5	22	○	●	○
3	2	23	●	○	○
3	2	24	○	○	○
3	5	25	●	○	○
3	2	26	○	○	○
3	2	27	○	○	●
3	5	28	○	●	○
3	2	29	●	○	○
3	2	30	○	○	○

1 2 3 4 5 6

● ○ ○ ○ ○ ○

Modulul de specializare